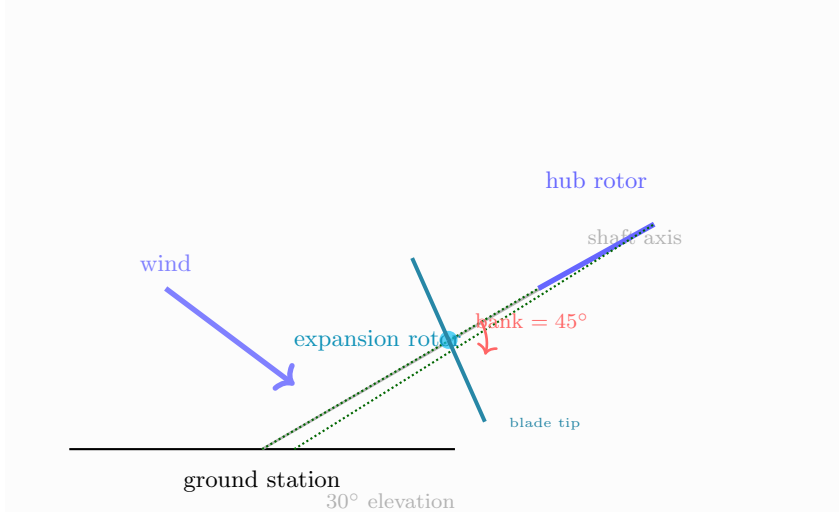
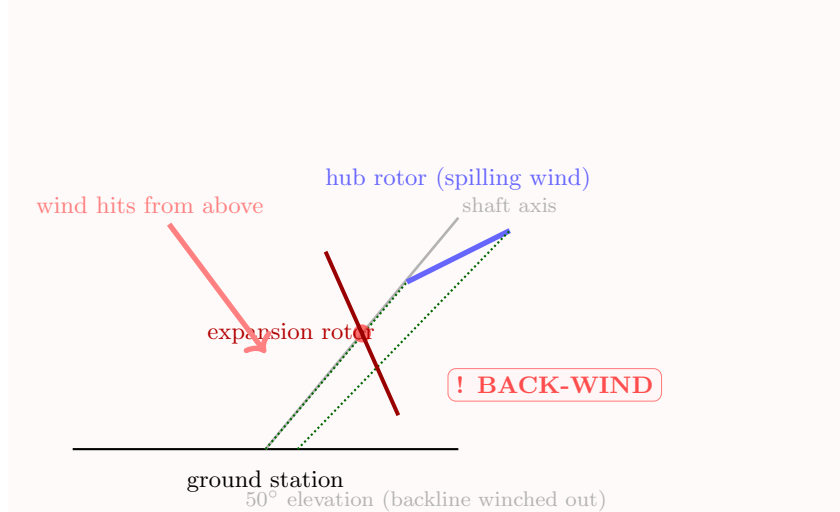


Expansion Rotor Bank Angle & Pitch Depower Risk

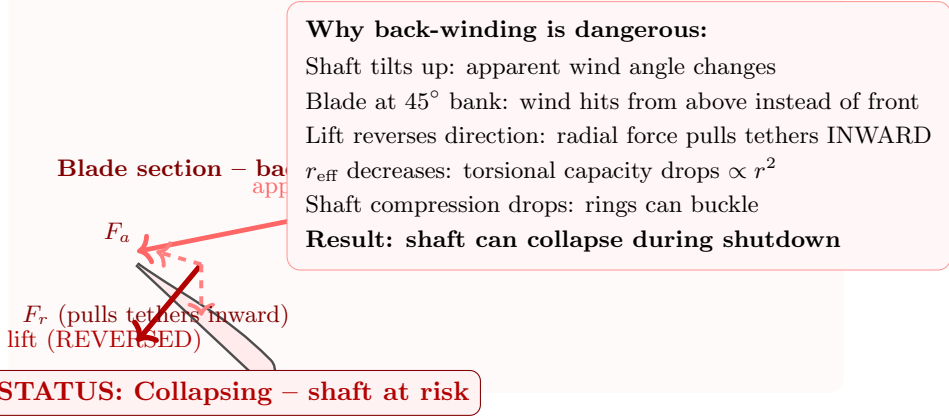
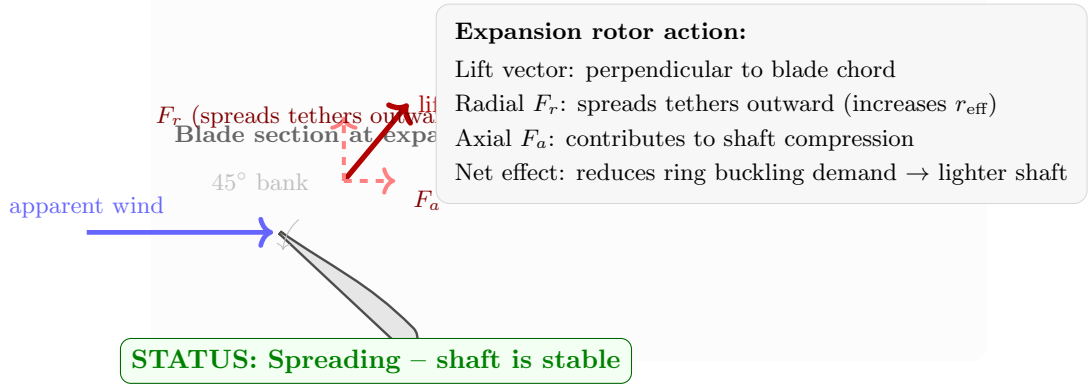
Normal Operation: Expansion Rotor Spreading



Pitch Depower: Back-Wind Risk



Pitch Depower



!! STATIC MODEL LIMITATION

The DE optimiser uses static structural sizing — it assumes F_r is always outward. It cannot penalise a design for failing during a transient pitch depower manoeuvre. **Dynamic ODE simulation is required.**

Mitigation options:

Lower bank angle (20–30°) · Symmetric/reflexed airfoils for reversed flow · Dynamic ODE validation · Conservative 20–30° for prototyping until validated